Week 5 Benchmark Report

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HPL

I built and ran HPL using MPI and OpenBLAS. I changed the `HPL.dat` parameters to:

- `N = 1000`
- `NB = 64`
- `P = 2`
- `Q = 2`

```
william@head:~/hpl/bin/linux$ cat HPL.dat
HPLinpack benchmark input file
Innovative Computing Laboratory, University of Tennessee
HPL.out      output file name (if any)
6            device out (6=stdout,7=stderr,file)
1            # of problems sizes (N)
1000         Ns
1            # of NBs
64           NBs
0            PMAP process mapping (0=Row-,1=Column-major)
1            # of process grids (P x Q)
2            Ps
2            Qs
16.0         threshold
3            # of panel fact
0 1 2        PFACTs (0=left, 1=Crout, 2=Right)
2            # of recursive stopping criterium
2 4          NBMINs (>= 1)
1            # of panels in recursion
2            NDIVs
3            # of recursive panel fact.
0 1 2        RFACTs (0=left, 1=Crout, 2=Right)
1            # of broadcast
0            BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1            # of lookahead depth
0            DEPTHs (>=0)
2            SWAP (0=bin-exch,1=long,2=mix)
64           swapping threshold
0            L1 in (0=transposed,1=no-transposed) form
0            U  in (0=transposed,1=no-transposed) form
1            Equilibration (0=no,1=yes)
8            memory alignment in double (> 0)
william@head:~/hpl/bin/linux$
```

I changed these values because the assignment requires different parameters from the presentation. `N` is the matrix size, `NB` is the block size, and `P × Q` is the process grid. Since I used 4 MPI processes, I chose a balanced `2 × 2` grid. The block size `64` is also within the recommended range from the slides.

Run command:

```
mpirun -np 4 --host head:2,work1:2 --map-by ppr:2:node ./xhpl > output.txt
```

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```
william@head:~/hpl/bin/linux$ tail -n 40 output.txt
================================================================================
T/V                N    NB     P     Q               Time                 Gflops
--------------------------------------------------------------------------------
WR00R2C4        1000    64     2     2               0.40             1.6676e+00
HPL_pdgesv() start time Fri Apr  3 15:22:03 2026

HPL_pdgesv() end time   Fri Apr  3 15:22:03 2026

--------------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=   5.20225868e-03 ...... PASSED
================================================================================
T/V                N    NB     P     Q               Time                 Gflops
--------------------------------------------------------------------------------
WR00R2R2        1000    64     2     2               0.43             1.5469e+00
HPL_pdgesv() start time Fri Apr  3 15:22:03 2026

HPL_pdgesv() end time   Fri Apr  3 15:22:04 2026

--------------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=   4.49164087e-03 ...... PASSED
================================================================================
T/V                N    NB     P     Q               Time                 Gflops
--------------------------------------------------------------------------------
WR00R2R4        1000    64     2     2               0.42             1.5943e+00
HPL_pdgesv() start time Fri Apr  3 15:22:04 2026

HPL_pdgesv() end time   Fri Apr  3 15:22:04 2026

--------------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=   4.23689109e-03 ...... PASSED
================================================================================

Finished     18 tests with the following results:
             18 tests completed and passed residual checks,
              0 tests completed and failed residual checks,
              0 tests skipped because of illegal input values.
--------------------------------------------------------------------------------

End of Tests.
================================================================================
william@head:~/hpl/bin/linux$
```

The run completed successfully and showed `PASSED`. Example Gflops results were:

- `1.6676e+00`
- `1.5469e+00`
- `1.5943e+00`

The final summary showed that all 18 tests passed.

HPCG

I built and ran HPCG with the following modified parameters:

- `56 56 56`
- `15`

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```
william@head:~/hpcg/build/bin$ cat hpcg.dat
HPCG benchmark input file
Sandia National Laboratories; University of Tennessee, Knoxville
56 56 56
15
william@head:~/hpcg/build/bin$
```

I changed them because the assignment requires values different from the presentation. These three numbers are the problem dimensions, and `15` is the runtime in seconds. I chose `56 56 56` because they are multiples of 8 and valid for the benchmark.

Run command:

```
mpirun -np 4 --host head:2,work1:2 --map-by ppr:2:node ./xhpcg
```

```
william@head:~/hpcg/build/bin$ tail -n 40 HPCG-Benchmark_3.1_2026-04-03_15-36-22.txt
Benchmark Time Summary::WAXPBY=0.140947
Benchmark Time Summary::SpMV=2.68084
Benchmark Time Summary::MG=14.5605
Benchmark Time Summary::Total=17.8957
Floating Point Operations Summary=
Floating Point Operations Summary::Raw DDOT=1.06072e+09
Floating Point Operations Summary::Raw WAXPBY=1.06072e+09
Floating Point Operations Summary::Raw SpMV=9.44433e+09
Floating Point Operations Summary::Raw MG=5.26475e+10
Floating Point Operations Summary::Total=6.42133e+10
Floating Point Operations Summary::Total with convergence overhead=6.42133e+10
GB/s Summary=
GB/s Summary::Raw Read B/W=22.1152
GB/s Summary::Raw Write B/W=5.11122
GB/s Summary::Raw Total B/W=27.2264
GB/s Summary::Total with convergence and optimization phase overhead=27.0414
GFLOP/s Summary=
GFLOP/s Summary::Raw DDOT=2.07215
GFLOP/s Summary::Raw WAXPBY=7.52565
GFLOP/s Summary::Raw SpMV=3.5229
GFLOP/s Summary::Raw MG=3.61577
GFLOP/s Summary::Raw Total=3.58821
GFLOP/s Summary::Total with convergence overhead=3.58821
GFLOP/s Summary::Total with convergence and optimization phase overhead=3.56383
User Optimization Overheads=
User Optimization Overheads::Optimization phase time (sec)=3.2e-07
User Optimization Overheads::Optimization phase time vs reference SpMV+MG time=4.63603e-06
DDOT Timing Variations=
DDOT Timing Variations::Min DDOT MPI_Allreduce time=0.199069
DDOT Timing Variations::Max DDOT MPI_Allreduce time=0.32629
DDOT Timing Variations::Avg DDOT MPI_Allreduce time=0.278221
Final Summary=
Final Summary::HPCG result is VALID with a GFLOP/s rating of=3.56383
Final Summary::HPCG 2.4 rating for historical reasons is=3.58821
Final Summary::Reference version of ComputeDotProduct used=Performance results are most likely suboptimal
Final Summary::Reference version of ComputeSPMV used=Performance results are most likely suboptimal
Final Summary::Reference version of ComputeMG used=Performance results are most likely suboptimal
Final Summary::Reference version of ComputeWAXPBY used=Performance results are most likely suboptimal
Final Summary::Results are valid but execution time (sec) is=17.8957
Final Summary::Official results execution time (sec) must be at least=1800
william@head:~/hpcg/build/bin$
```

The result file showed:

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- `HPCG result is VALID`
- GFLOP/s rating = `3.56383`

STREAM

STREAM measures main memory bandwidth. I changed the array size from the presentation example to:

- `STREAM_ARRAY_SIZE = 60000000`

```
william@head:~/STREAM$ cat output.txt
-------------------------------------------------------------
STREAM version $Revision: 5.10 $
-------------------------------------------------------------
This system uses 8 bytes per array element.
-------------------------------------------------------------
Array size = 60000000 (elements), Offset = 0 (elements)
Memory per array = 457.8 MiB (= 0.4 GiB).
Total memory required = 1373.3 MiB (= 1.3 GiB).
Each kernel will be executed 30 times.
 The *best* time for each kernel (excluding the first iteration)
 will be used to compute the reported bandwidth.
-------------------------------------------------------------
Number of Threads requested = 2
Number of Threads counted = 2
-------------------------------------------------------------
Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 26960 microseconds.
   (= 26960 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.
-------------------------------------------------------------
WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.
-------------------------------------------------------------
Function    Best Rate MB/s  Avg time     Min time     Max time
Copy:           34579.7     0.028228     0.027762     0.029138
Scale:          20915.4     0.046033     0.045899     0.046480
Add:            22085.9     0.065691     0.065200     0.066718
Triad:          22130.8     0.065510     0.065068     0.067840
-------------------------------------------------------------
Solution Validates: avg error less than 1.000000e-13 on all three arrays
-------------------------------------------------------------
william@head:~/STREAM$
```

I changed it because the assignment requires a different parameter. The array size is related to hardware because if the array is too small, the data may stay in cache. A large array is better for measuring real main-memory bandwidth.

Compile command:

```
make clean
make CC=gcc CFLAGS="-O3 -march=native -fopenmp -DSTREAM_ARRAY_SIZE=60000000 -
DNTIMES=30 -DSTREAM_TYPE=double" stream_c.exe
```

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Run command:

```
./stream_c.exe
```

Results:

- Copy: `34579.7 MB/s`
- Scale: `20915.4 MB/s`
- Add: `22085.9 MB/s`
- Triad: `22130.8 MB/s`

The output ended with `Solution Validates`.

OSU

I used OSU to compare MPI communication on the same node and on different nodes.

Commands:

```
mpirun -np 2 --host head:2 ./osu_latency
mpirun -np 2 --host head:2 ./osu_bw

mpirun -np 2 --host head:1,work1:1 ./osu_latency
mpirun -np 2 --host head:1,work1:1 ./osu_bw
```

Same node (`head:2`):

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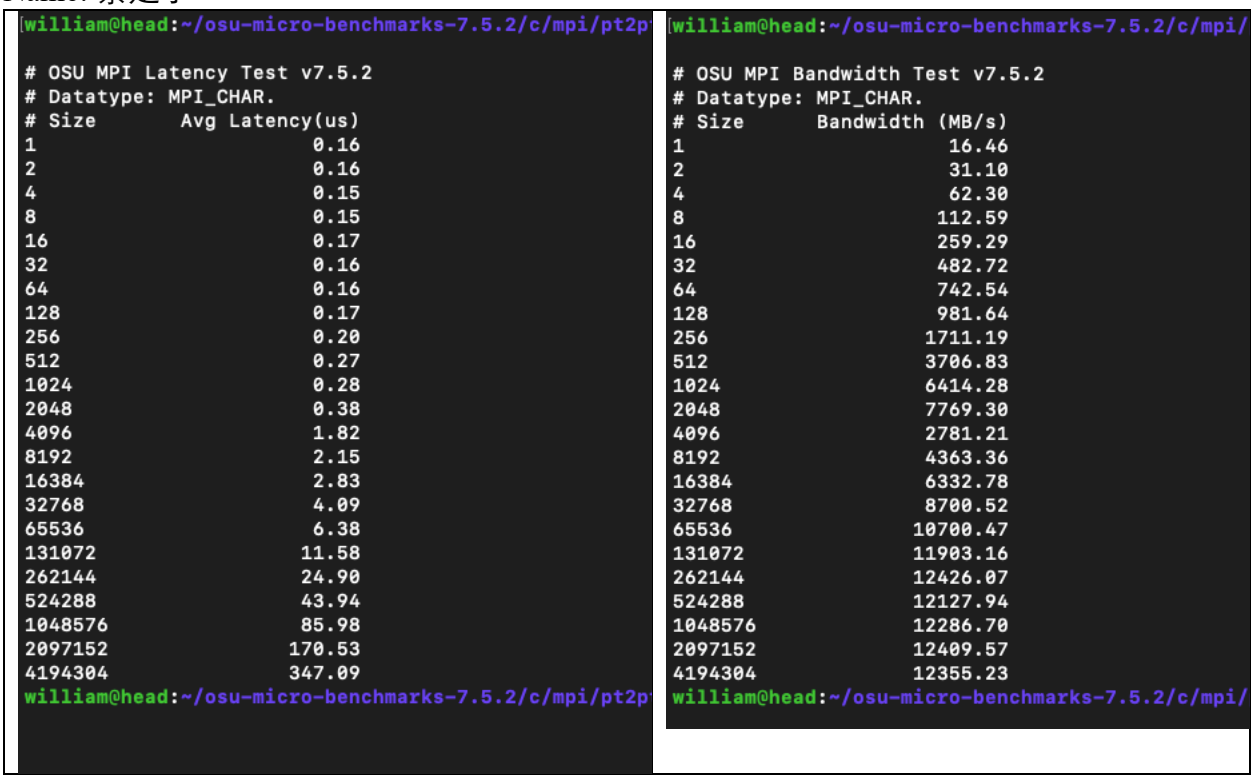

```
william@head:~/osu-micro-benchmarks-7.5.2/c/mpi/pt2p

# OSU MPI Latency Test v7.5.2
# Datatype: MPI_CHAR.
# Size       Avg Latency(us)
1                       0.16
2                       0.16
4                       0.15
8                       0.15
16                      0.17
32                      0.16
64                      0.16
128                     0.17
256                     0.20
512                     0.27
1024                    0.28
2048                    0.38
4096                    1.82
8192                    2.15
16384                   2.83
32768                   4.09
65536                   6.38
131072                 11.58
262144                 24.90
524288                 43.94
1048576                85.98
2097152               170.53
4194304               347.09
william@head:~/osu-micro-benchmarks-7.5.2/c/mpi/pt2p
```

```
william@head:~/osu-micro-benchmarks-7.5.2/c/mpi/

# OSU MPI Bandwidth Test v7.5.2
# Datatype: MPI_CHAR.
# Size      Bandwidth (MB/s)
1                      16.46
2                      31.10
4                      62.30
8                     112.59
16                    259.29
32                    482.72
64                    742.54
128                   981.64
256                  1711.19
512                  3706.83
1024                 6414.28
2048                 7769.30
4096                 2781.21
8192                 4363.36
16384                6332.78
32768                8700.52
65536               10700.47
131072              11903.16
262144              12426.07
524288              12127.94
1048576             12286.70
2097152             12409.57
4194304             12355.23
william@head:~/osu-micro-benchmarks-7.5.2/c/mpi/
```

- 1-byte latency: `0.16 us`
- peak bandwidth: `12426.07 MB/s`
- 4 MiB latency: `347.09 us`

Different nodes (`head:1,work1:1`):

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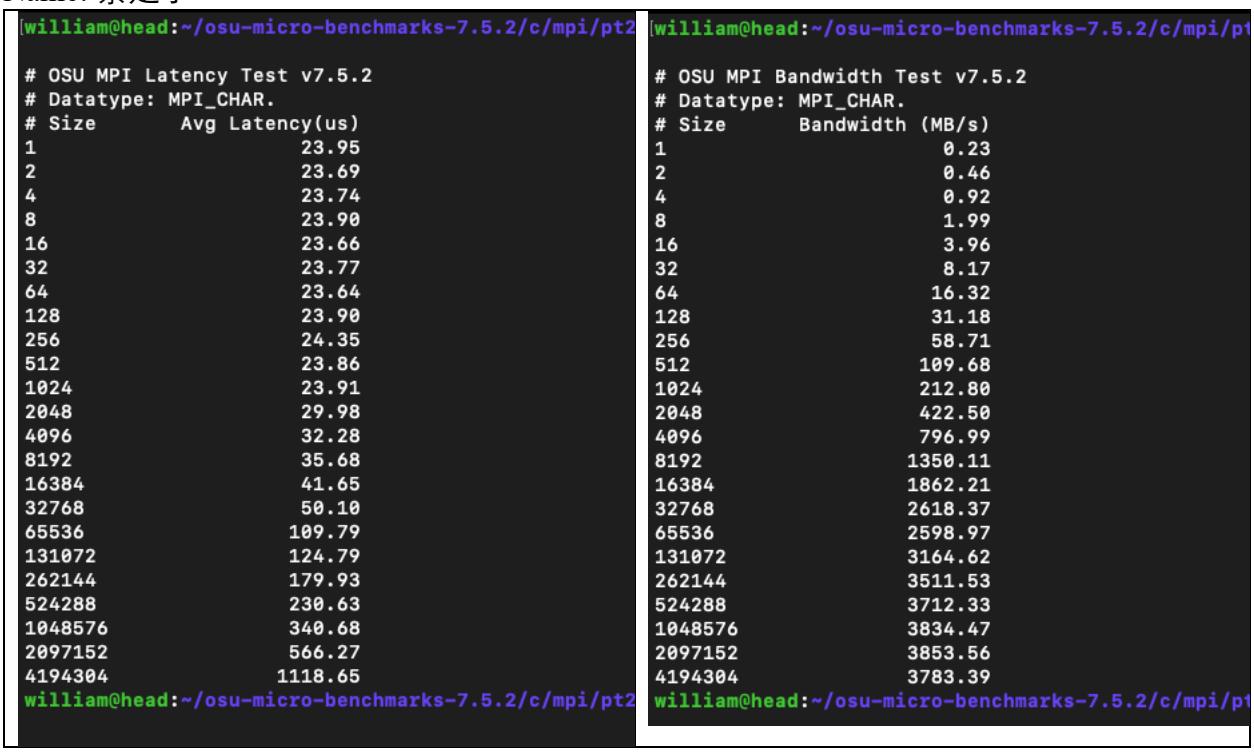

- 1-byte latency: `23.95 us`
- peak bandwidth: `3853.56 MB/s`
- 4 MiB latency: `1118.65 us`

The same-node result is faster because communication stays inside one machine. The different-node result is slower because data has to go through the network between nodes.

IOR

I used IOR with parameters different from the presentation:

```
mpirun -np 2 ./src/ior \
  -w -r \
  -t 2m \
  -b 256m \
  -F \
  -o testfile
```

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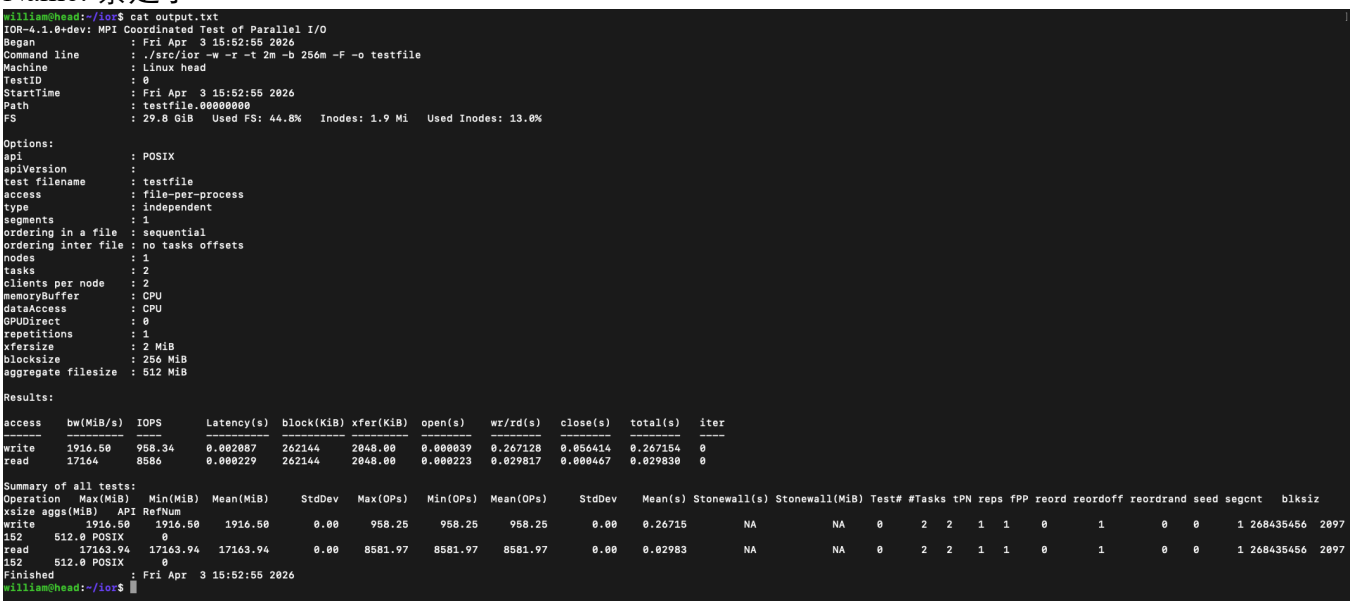

I changed `-t` and `-b` because the assignment requires different parameters. Here, `-t 2m` means 2 MiB transfer size, and `-b 256m` means each process writes a 256 MiB block. `-F` means file-per-process.

Results:

- write bandwidth: `1916.50 MiB/s`
- read bandwidth: `17164 MiB/s`

The read speed is much higher than the write speed, which likely shows the effect of **page cache**. This means some read data may have been served from memory instead of storage.

Conclusion

In this assignment, I tested HPL, HPCG, STREAM, OSU, and IOR to measure compute, memory, communication, and storage performance. By changing the parameters from the presentation examples, I was able to observe how each benchmark behaves under different settings. The results also show that different parts of the system can become the main bottleneck depending on the workload.